CHAPTER 5

The KuiSCIMA Dataset

In this section, we investigate Research Question (RQ2): How can we efficiently build a digital corpus of *Baishidaoren Gequ*?

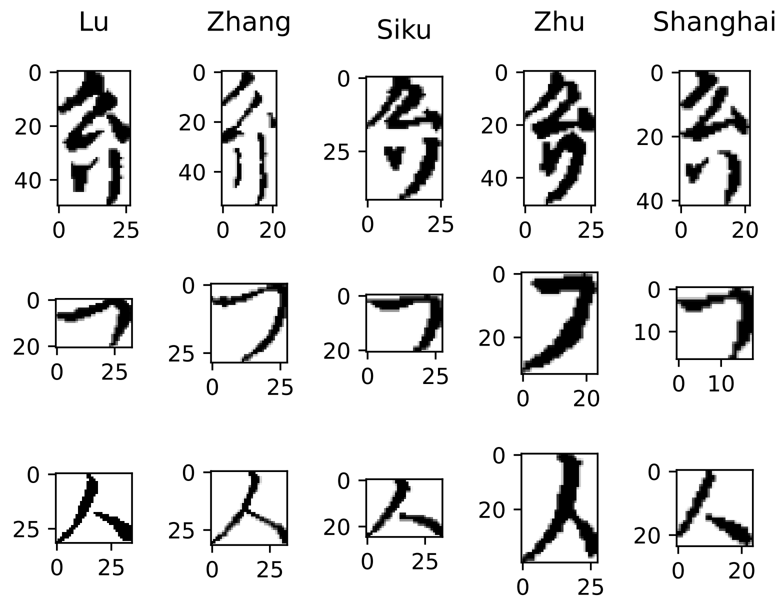
The result of this section is the KuiSCIMA (short for *Jiang Kui Score Images for Musicological Analysis*) dataset, consisting of *Baishidaoren Gequ* in five different editions. KuiSCIMA is the first publicly available OMR dataset of Chinese music containing *suzipu* notation. Its name is based on the CVC-MUSCIMA dataset (Fornés et al., 2012).

We are going to see which historical editions are included in the KuiSCIMA corpus, how they relate to its five optical symbolic digital editions and two purely symbolic editions, and how the images were collected and preprocessed. Then, we explain the annotation process, including the challenges and what choices were made in case of ambiguous instances. In addition, we present some numbers and figures regarding dataset composition.

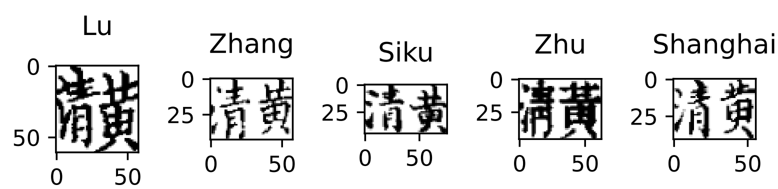
5.1 Included Editions

The KuiSCIMA dataset focuses on Jiang Kui's collection *Baishidaoren Gequ*, which was originally published in 1202. It contains 109 pieces in total, from which 28 are endowed with musical notations. To ensure a variety of writing styles in this rather small set of pieces, five different woodblock print editions have been included, as described in Section 3.5: the Lu edition from 1743 (Jiang, 1202/2011), the Zhang edition from 1749 (Jiang, 1202/2019b), the *Siku Quanshu* edition from the end of the 18$^{\text{th}}$ century (Jiang, 1202/n.d.), the Zhu edition from 1913 (Jiang, 1202/1913), and the Shanghai MS (where *MS* is short for *manuscript*) from around 1736 (Jiang, 1202/2019a). The first four editions are woodblock prints, while the latter is a manuscript.

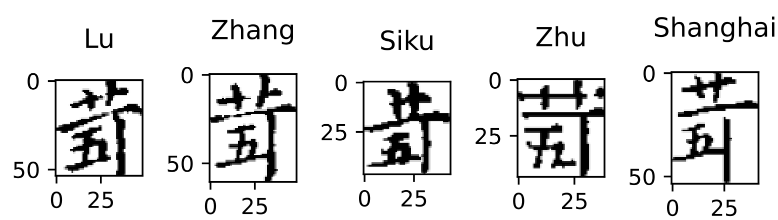
KuiSCIMA features symbol-level content and position annotations for both text and notation characters together with the binarized source images. A small sample of the notations included in the corpus is found in Figure 5.1.



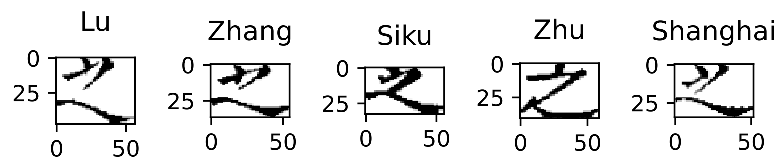
(a) *Suzipu* styles



(b) *Lülüpu* styles

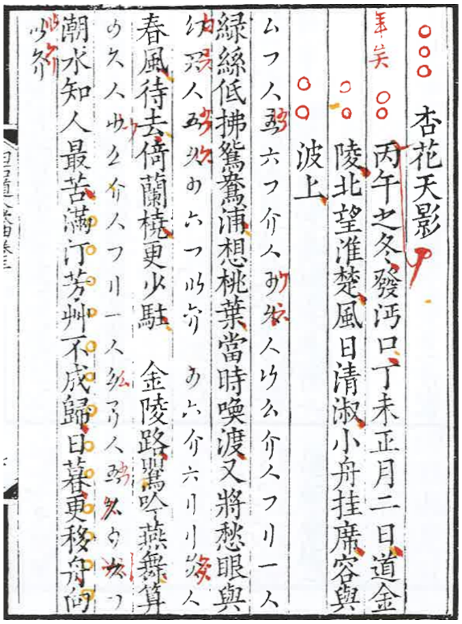


(c) *Jianzipu* styles



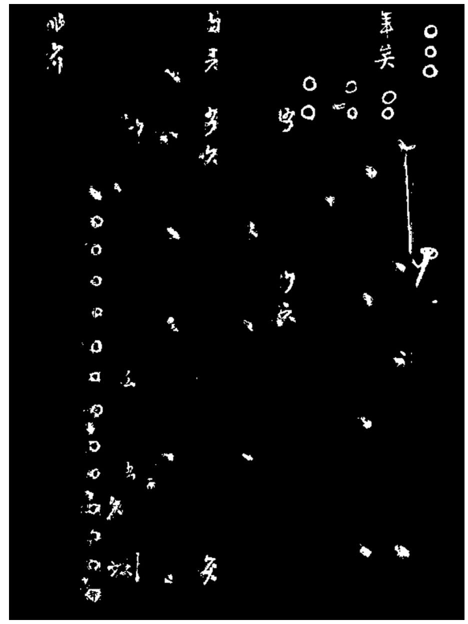
(d) Text styles

Figure 5.1: Here, for each of the four annotation types present in the dataset, some examples of notations are visualized to highlight the differences in writing styles.

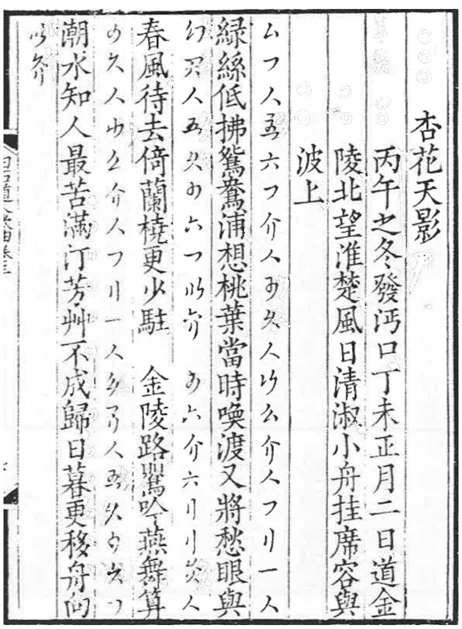


杏花天影
丙午之冬發沔口丁未正月二日道金
陵北望淮楚風日清淑小舟挂席容與
波上
綠絲低拂鴛鴦浦想桃葉當時喚渡又將愁眼與
春風待去倚蘭橈更少駐 金陵路鶯吟燕舞算
潮水知人最苦滿汀芳艸不成歸日暮更移舟向

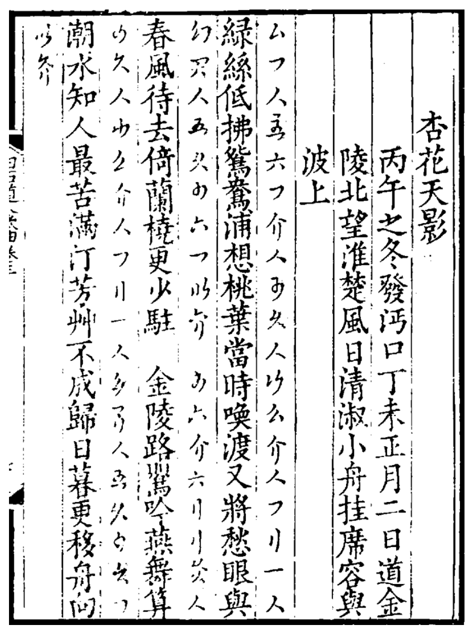
(a) The warped and cropped page.



(b) Masking the pixels in HSV space: saturation between [35, 255], and value between [155, 235].



(c) Binary OR operation between the original image and the masked pixels, leading to a removal of the colors from the image.



(d) Result after applying Otsu's method.

Figure 5.2: Visualization of the individual preprocessing steps for images of the Zhang edition, the page containing *Xinghuatianying* is chosen as an example.

5.2 Raw Image Processing

The 1743 Lu edition (Jiang, 1202/2011) and the *Siku Quanshu* edition (Jiang, 1202/n.d.) are available online on the Chinese Text Project (Sturgeon, 2011) and have been manually downloaded. The images are in PNG format, already binarized, and resized to a width of 800 pixel and variable height. In the *Siku Quanshu* edition of the piece *Nishangzhongxu Diyi*, the first four columns of the piece (which contain the title and a part of the preface) contain artifacts due to a suboptimal digitization of a red seal.

The other three editions must be preprocessed to be used for the dataset:

1. The 1913 Zhu edition (Jiang, 1202/1913) is scanned, each of the double-page images is split into halves, rotated to fit a rectangle frame if too skewed, cropped to the area of the woodblock print, converted to grayscale, binarized using OpenCV's (Bradski, 2000) implementation of Otsu's method, resized to 800 pixels in width using the `cv2.INTER_AREA` flag, and then binary thresholded with a threshold of 235. With the OpenCV function `connectedComponentsWithStats`, small noisy areas are cleaned up by removing all connected components with an area less than or equal to 10.

2. For the Shanghai MS and the 1749 Zhang edition (Jiang, 1202/2019b), a slightly different method is used: The individual pages are scanned, warped using the corrective perspective tool in GIMP if too skewed, and cropped. Since the Zhang edition contains remarks in black, yellow and red, the yellow and red ones are removed before the grayscale conversion by extracting and replacing the colored parts with white in the HSV color space. This color removal is not done for the Shanghai MS, since no colored remarks are present. Then, the images are resized to 1600 pixels in width, Otsu's method is applied, and the images are individually split into halves. The individual stages of preprocessing are illustrated in Figure 5.2.

5.3 Annotation Process

Using the annotation GUI introduced in Section 4.5.1, for each of the 109 pieces, the pages (which are already preprocessed as described in Section 5.2) containing a single piece are selected. This includes the title, possibly the mode information and the preface, and the lyrics with the notation symbols (if present).

Then, using the HRCenterNet algorithm by Tang et al. (2020), for each page a segmentation of the image is performed individually, yielding a set of rectangles, each of which are supposed to correspond to the bounding box of a character in the image. Using this preliminary segmentation, the boxes not belonging to the piece are removed, and the boxes are checked individually and corrected if necessary with the GUI, so that each bounding box encompasses the character adequately. Afterwards, the bounding boxes are minimized, i.e., there are no empty rectangles at each side of the bounding box.

Additionally, the blank spaces in both the lyrics and music columns separating the stanzas are represented by a bounding box containing an empty string as annotation, in order to encode the typographical layout of each edition as closely as possible. In the next step, each box is assigned a label (see Table 4.4) according to its function in the document: For example, boxes containing a character of the piece's title are marked with the `Title` label, the notation symbols with the `Music` label etc.

After this, each box is annotated such that the annotation corresponds with the semantic meaning of the character. In case of the text-based boxes, each annotation is a string containing up to a single Chinese character; and in case of the music-based boxes, each annotation is the digital representation of the single notation symbol as introduced in Chapter 4. Finally, the mode information is annotated and the piece is converted to the JSON representation as described in Section 4.1.

5.4 Annotation Remarks

There are three reasons why the annotations in this corpus need further remarks:

1. All the editions in this corpus are handwritten, since they are either manuscripts or woodblock prints (which is a process involving manually carving out the shapes of each individual character). Similar to modern prints, errors when copying the source material may occur, and errors already included in the source material itself may have been copied.

2. When printing, some parts of the woodblock may have worn off, or the ink was not distributed evenly, causing artifacts in the final print (Sturgeon, 2018, p. 18).

3. The editions were copied from the Tao MS in the Qing dynasty (1644–1912), at a time where the *suzipu* notation was already out of use and *ci* poetry had long turned from a musical practice into a mostly literary genre (Y. Yang, 2019, p. 42). As such, the copyists needed to copy the symbols without background knowledge about them, thus increasing the chance of errors.

These factors result in anomalous shapes or misprints of the text or the notation symbols in *Baishidaoren Gequ*, such that the meaning is unclear and needs further interpretation. Some of such erroneous/unclear symbol instances are common to parallel places in multiple editions of this corpus, while others are restricted to individual editions.

The annotations in this corpus are made to preserve the original shapes of the notation symbols as closely as possible, but restricted to the framework given in Section 3.3. Unclear notation symbols are marked with the `excluded_from_dataset` tag, and the dataset annotation is selected to be as close as possible to the instance's original appearance. For the text characters, the most similar common variant is chosen, while in ambiguous cases, the character is annotated as described in (Y. Yang, 2019).

5.4.1 Symbolic Corpus

The folder `symbolic_dataset/01_symbolic_edition`[1] contains the basic edition of all 109 pieces. Regarding the *lülüpu* notations, in cases of inter-edition disagreements, the

[1]https://github.com/SuziAI/KuiSCIMA/tree/v2.0/KuiSCIMA/symbolic_dataset/01_symbolic_edition

annotations are chosen to reflect the variant that appears most often in the parallel places of all five editions. The *jianzipu* piece in this edition is based on the transnotation by Thompson (n.d.-a), where the original notation features are marked with a square. Regarding the *suzipu* notations, the edition of Wu (2013, pp. 312–328) is the basis for this symbolic-only corpus, not only with respect to the notation, but also regarding the layout. For all 109 pieces, the punctuation marks and the lyrics (with normalization of variant Chinese characters to a more consistent text) are taken from the new critical edition of *Baishidaoren Gequ* by Y. Yang (2019, pp. 89–187).

Therefore, this edition follows the historical editions of *Baishidaoren Gequ* closely, including pitch symbols which lie outside the pieces' modes. For certain applications such as music generation, it is beneficial to normalize the pitches such that only pitches which are in accordance with the pieces' modes are present in the data, a property which is introduced as *modal consistency* in Section 7.1. For the *jianzipu* piece, the corrections from Thompson (n.d.-a) which are marked by a circle are used. This edition, called *normalized edition* is provided in the folder `symbolic_dataset/02_normalized_edition`[2], and it only contains the 28 pieces containing musical notations.

The list of individual annotation remarks for the symbolic corpus is in Appendix B.1.

5.4.2 Optical Symbolic Corpus

As opposed to the symbolic corpus from above, the optical symbolic corpus contains symbolic notational and textual information together with the bounding boxes which annotate the photographic reproductions of the historical editions. This dataset is the basis for the development and evaluation of OMR methods acting on different kinds of notations.

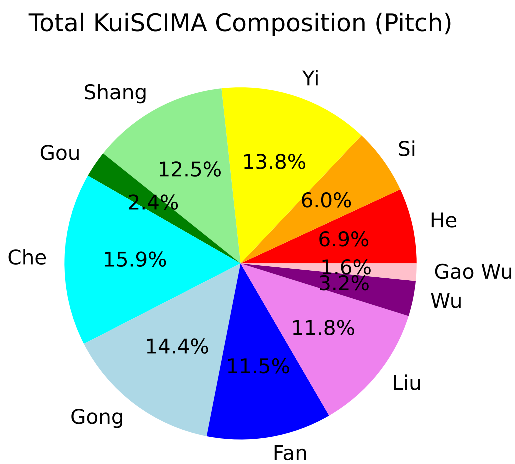
In KuiSCIMA `v2.0`, there are 11,529 musical annotations in the optical symbolic dataset, from which 7,212 (62.6%) are *suzipu*, 3,385 (29.4%) are *lülüpu*, and 708 (6.1%) are *jianzipu* annotations. The remaining 224 (1.9%) are empty notation boxes, indicating the absence for structuring purposes inside the pieces, such as the separation of stanzas.

The folder `KuiSCIMA/optical_symbolic_dataset`[3] contains all individual editions in folders named after the contained edition, i.e., `01_lu_edition`, `02_zhang_edition`, `03_siku_quanshu_edition`, `04_zhu_edition`, and `05_shanghai_manuscript`. Inside each of these folders are two subfolders, one for the photographic reproductions called `images`, and one for the JSON files called `json`.

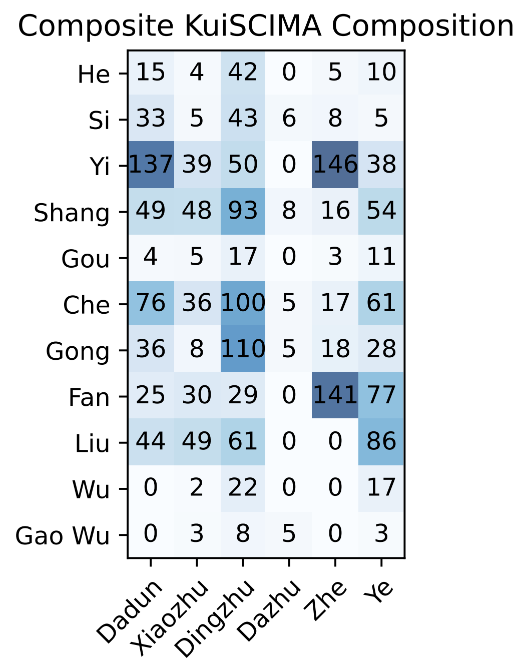
The instances in the optical symbolic dataset feature a highly imbalanced class distribution with regards to *suzipu* and *jianzipu* notations as visualized in Figure 5.3.

In addition, the task of identifying the composite notations is difficult, since the individual strokes of a symbol may be separated or connected. As opposed to Chinese character OCR, any *suzipu* symbol has at most two components. However, the degree of ambiguity in *suzipu* is much higher due to the small set of components and their similarity to each other. This ambiguity is shown in Figure 5.4.

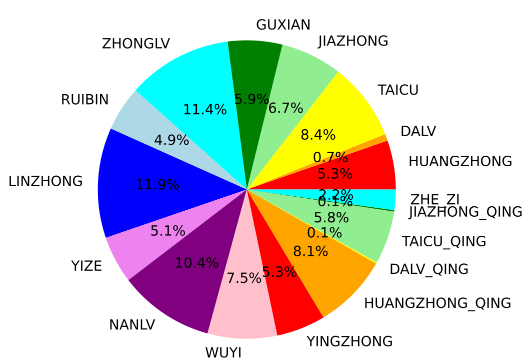
[2]https://github.com/SuziAI/KuiSCIMA/tree/v2.0/KuiSCIMA/symbolic_dataset/02_normalized_edition
[3]https://github.com/SuziAI/KuiSCIMA/tree/v2.0/KuiSCIMA/optical_symbolic_dataset



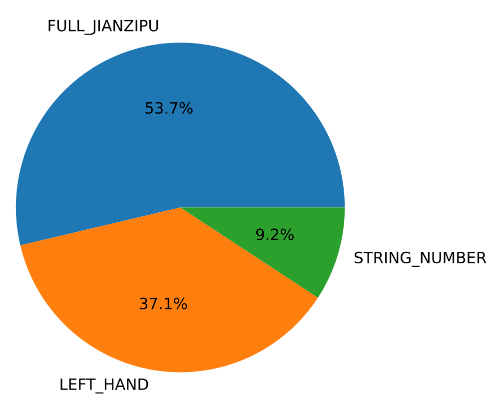
(a) The *suzipu* total pitch distribution indicates that the pitches are highly imbalanced.



(b) The *suzipu* composite notation distribution illustrates that the pitch and secondary pairs are locally concentrated: only 66 out of 77 possible combinations occur in the dataset.



(c) *Lülüpu* notation class distribution



(d) *Jianzipu* class distribution

Figure 5.3: The balance of the KuiSCIMA dataset. In (a) and (b), the distribution of the *suzipu* notations is illustrated, while in (c) and (d) the balance of the *lülüpu* and *jianzipu* classes is shown.

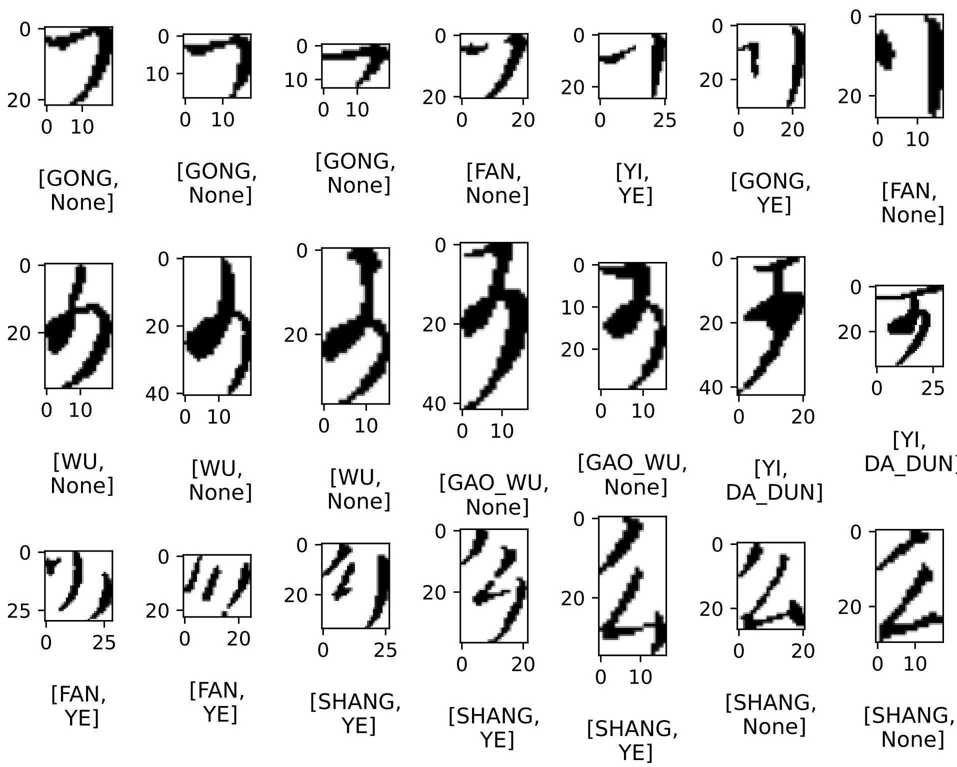


Figure 5.4: In each row, examples of similar-looking *suzipu* notations in the Shanghai MS are visualized.

In some pieces, the Shanghai MS features manual corrections regarding the stanza separation, this is indicated by a vertical stroke deleting the original stanza separation, and a small circle where the separation is placed instead. This occurs in seven pieces, namely *Nishangzhongxu Diyi*, *Meiwu* 眉嫵, *Yuexiadi* 月下笛, *Faquxianxianyin* 法曲獻仙吟, *Changtingyuanman*, *Danhuangliu*, and *Jueshao*. The corrected stanza separations agree with the separations in all other four editions. This original placement of the stanza divisions is hypothesized to have been present in the first edition of *Baishidaoren Gequ* and may indicate a special musical practice (Y. Yang, 2019, p. 55). Therefore, the original variant is preserved in the digitization of the Shanghai MS.

The list of individual annotation remarks for the optical symbolic corpus is found in Appendix B.2.

5.5 Summary

In this chapter, we learned about the optical symbolic corpus of the KuiSCIMA dataset, with its five historical editions of *Baishidaoren Gequ* that are endowed with position-encoded symbolic representations and the image preprocessing and artifact removal pipeline; and about the origins of the two editions in the purely symbolic corpus.

For both corpora, the annotation process was described thoroughly, and reasons for why some annotation choices need further explanation are stated (e.g., unfamiliarity of the scribes with the musical notations, misprints, copying or printing errors). The full list of annotation remarks is found in Appendix B.1 for the purely symbolic corpus, and in Appendix B.2 for the optical symbolic corpus.

In addition, the label distributions of the 11,529 musical annotations in the optical symbolic dataset are discussed and visualized, and some continuous transformations in the image space yielding different discrete annotations are shown. The result is the KuiSCIMA `v2.0` dataset with 62,665 manually checked annotations across 511 pages.

Using this dataset, we can now turn our attention to Research Question (RQ3) and develop different OMR methods (such as raw image clustering in Section 6.1, calibrated CNNs in Sections 6.3 and 6.4, and feature-based UMAP clustering in Section 6.6) with a special focus on *suzipu* and *lülüpu* notations. We are also going to compare the methods against out-of-the-box baselines in the *lülüpu* case (Section 6.4.4) and against a baseline of naive human users (Section 6.7) for *suzipu* recognition.

This and a description of the challenges are found next in Chapter 6.